

Chapter 05 Energy

Forms of Energy

- Mechanical
- Chemical
- Electromagnetic
- Nuclear

Work-Energy and Force

$$W = F \cos \theta \Delta x$$

Work is scalar quantity.

Kinetic Energy

$$KE = \frac{1}{2}mv^2$$

and

$$W_{net} = KE_f - KE_i = \Delta KE$$

Types of Forces

Two general classes of forces:

- conservative: Work and energy associated with the force are only related to initial and final position.
- Nonconservative: Related to path.

Potential Energy

Associated with the position of the object within some system. It is property of the system, not the object.

For every conservative force, a potential energy function can be found.

Work-Energy Theorem

$$W_{nc} = (KE_f - KE_i) + (PE_f - PE_i)$$

Conservation of Energy

Total mechanical energy is the sum of the kinetic and potential energies in the system

$$E_i = E_f$$
$$KE_i + PE_i = KE_f + PE_f$$

Springs

Involves **spring constant**, k . And Hooke's Law gives

$$\vec{F} = -k\vec{x}$$

The F is the **restoring force**, which is conservative force.

Thus it corresponds to **elastic potential energy**. And

$$PE_s = \frac{1}{2}kx^2$$

Nonconservative Forces with Energy Considerations

When nonconservative force are present, the total **mechanical energy** is **not constant**. And the work done by all nonconservative forces equals to the change of mechanical energy:

$$W_{nc} = \Delta E$$

Power-Energy Transfer

Power is the **rate** where the force works on an object, defined as

$$P_{avg} = \frac{W}{\Delta t}$$

And instantaneous power:

$$P = \frac{dW}{dt}$$

or

$$P = \vec{F} \cdot \vec{v}$$

Center of Mass

The **point** in the body at which all the mass can be considered to be concentrated. The **potential energy** is related to the position of center of mass.

For many particles in two-dimensional plane, the position vector

$$\vec{r}_i = x_i\vec{i} + y_i\vec{j} + z_i\vec{k}$$

the position of the center of mass of a system of particles is:

$$\vec{r}_{com} = x_{com}\vec{i} + y_{com}\vec{j} + z_{com}\vec{k}$$

for x_{com} ,

$$x_{com} = \frac{1}{M} \sum_{i=1}^n m_i x_i$$

where $M = m_1 + m_2 + \cdots + m_n$

Thus

$$\vec{r}_{com} = \frac{1}{M} \sum_{i=1}^n m_i \vec{r}_i$$

Work Done by Varying Forces

Equal to the area under the graph of F vs. x .

$$W = \int_{x_i}^{x_f} F(x) dx$$

Work-Kinetic Energy Theorem with a Variable Force

$$W = \int_{v_i}^{v_f} mv dv = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$$